

Jyotishman Pathak

Jorhat, Assam, India (Remote – Global)

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SUMMARY

Backend engineer specializing in real-time systems and distributed infrastructure. Built production-grade applications using Node.js, Redis, and WebRTC with focus on caching strategies, NAT traversal, and production deployment. Ships end-to-end features: from React hooks to database migrations to Dockerized services. Seeking remote backend/real-time engineering roles.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, SQL, Python
- **Backend:** Node.js, Express, Fastify, Socket.io, WebRTC, Redis (Streams, Pub/Sub, Lua), PostgreSQL, Prisma
- **Frontend:** Next.js (App Router), React, Tailwind CSS, Zod
- **Infrastructure:** Docker, GitHub Actions, AWS (S3, IAM, EC2 basics), Upstash, Render, Vercel
- **Concepts:** Caching patterns, distributed locks, ICE/STUN/TURN, event-driven architecture, API design, load testing (k6/autocannon)

PROJECTS

GhostMesh – Zero-Knowledge P2P Collaboration Platform *Next.js, Node.js, Redis (Upstash), WebRTC*
Browser-based P2P video/chat/file-transfer with ephemeral signaling and privacy-first auth

- Implemented ephemeral Redis signaling with 60s TTL self-destruct; reduced server storage footprint by 99%
- Built ICE candidate queuing via LPUSH/LRANGE; handled symmetric NAT fallback with TURN relay
- Reduced signaling bandwidth by 78% by killing polling intervals when `dataChannel.readyState === "open"`
- Containerized with Docker, deployed via Vercel + Upstash; added GitHub Actions CI and k6 load tests

TicketForge – High-Concurrency Event Ticketing Backend *Node.js, Redis, Fastify, ioredis, Lua, Docker, k6*
Production-grade ticketing API handling cache-aside, distributed locks, and Stream-based order queues

- Prevented overselling via WATCH/MULTI transactions with optimistic locking retry logic
- Implemented sliding-window rate limiter using Lua scripting (atomic server-side execution)
- Replaced list-based queue with Redis Streams + consumer groups; added XACK tracking for stuck jobs
- Benchmarked pipelining: 3.2x throughput improvement on bulk event seeding (1000 events)

EXPERIENCE

Backend Engineer Intern

May 2024 – July 2024

Bluestack Fintech, Mumbai

Node.js, NestJS, PostgreSQL, Redis

- Engineered financial data service reducing API latency by 40% via query optimization and PostgreSQL indexing
- Implemented Redis cache-aside layer, decreasing primary database read load by 70%
- Designed token-bucket rate limiter using Redis Lua scripting to protect downstream services

EDUCATION

Bachelor of Engineering in Electronics and Instrumentation

2023 – 2027

Jorhat Institute of Science and Technology, Jorhat, Assam, India

Relevant Coursework: Data Structures, Algorithms, Database Systems, Computer Networks

ADDITIONAL

- **DSA:** 150+ LeetCode problems (Arrays, Hashing, Two Pointers, Trees, BFS/DFS)
- **Writing:** Technical blog posts on Redis caching patterns and WebRTC debugging and web related topics